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Palaeomagnetic study of Mesozoic basalts from Scania, southernmost Sweden

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SUMMARY

The Mesozoic basalt necks of central Scania, southernmost Sweden, yield dual polarity palaeomagnetic directions: decl. 33°, incl. 66°, $N = 15$ and decl. 225°, incl. −73°, $N = 6$. The combined mean with the negative directions reversed is decl. 35°, incl. 68°, $N = 21$. The corresponding palaeopole is situated at lat. 69°N, 103°E. This pole corresponds with an Early Jurassic pole for Europe and with Early Jurassic poles for North America after closing of the Atlantic. The Early Jurassic age is in contrast with K–Ar datings, ranging between 171 and 81 Ma, but is in agreement with datings based on microfossils found in tuffaceous layers connected with the basalts.

Key words: basalt, Early Jurassic, palaeomagnetism, Sweden, titanomagnetite.

INTRODUCTION

Our understanding of the Mesozoic Apparent Polar Wander Path (APWP) for Eurasia has been hampered by the paucity of well dated Mesozoic rocks with reliable palaeomagnetic records. Recently a new palaeopole from the Late Jurassic/Early Cretaceous Hinlopenstretet dolerites (144 Ma) was reported from Svalbard at the northwestern corner of the Eurasian plate (Halvorsen 1989). The palaeopole at 66°N, 200°E compares well with the palaeopole from the upper Morrison Formation (145 Ma) on the North American craton, after compensating for the post-144 Ma plate separation using the rotation data from Frei & Cox (1987). These data and other contemporary palaeomagnetic poles will be compared with results from a set of Mesozoic basalts from Scania, southernmost Sweden, lat. 56.0°N, long. 13.5°E.

A previous investigation of 11 Scanian basalt sites was undertaken by Bylund (1973) in an unpublished study. Further samples from these sites and 12 more sites are included in this paper. Also included in Table 1 and the discussion are data from four sites published by Schweitzer (1975).

GENERAL GEOLOGY

The province of Scania forms the southernmost part of the Scandinavian Peninsula (Fig. 1). The geology of the province is determined by its position on the border between the stable Pre-Cambrian Baltic Shield and the fragmented sedimentary platform of western Europe. This border zone, the Fennoscandian Border Zone (FBZ), forms the western part of the Tornquist–Tesseyre lineament which stretches from the Oslo–Bamble–Horn Graben in the North Sea along the Swedish west coast, across Scania and the Baltic through Poland and the Ukraine to the Black Sea. In Scania the FBZ trends roughly NW–SE and is approximately 60 km wide. In addition, the Protogine Zone (PZ), is a circa 30 km-wide NNE-trending lineament which transects the province (Fig. 1). It separates the Pre-Cambrian southwest Scandinavian Gneiss Complex which forms the southwestern part of the Baltic Shield from its major Pre-Cambrian Svecofennian part. This position in a border zone transected by major lineaments creates a mosiac pattern in the Scanian geology. For references on the FBZ and the PZ see Bergström *et al.* (1982) and Andreasson & Rodhe (1990).

The northern Pre-Cambrian part of Scania consists of gneisses and granites, while the southern part is made up of Palaeozoic and Mesozoic sediments with single outliers of the Pre-Cambrian crystalline basement forming the Scanian horsts. Movements in the PZ were mainly restricted to the Pre-Cambrian (Patchett & Bylund 1977; Andreasson & Rodhe 1990). The FBZ has been active in Palaeozoic and Mesozoic time (e.g. Bölau 1966; Bergström *et al.* 1982; Norling & Bergström 1987). The dominating NW–SE trends in the Scanian geology are exemplified by the basement horsts and an extensive swarm of Late Carboniferous–Early Permian dolerites. In Kimmerian time block movements were prominent and the characteristic Scanian horst

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structures became even more pronounced as fault tectonics dominated. Bergström *et al.* (1982) have presented a comprehensive summary of the Scanian geology, and Norling & Bergström (1987) have reviewed the tectonic history.

The Scanian basalts occur in an elongated area in central Scania. They outcrop as isolated volcanic plugs from which the original lava has been eroded. About seventy necks are reported (Bergström *et al.* 1982), although recent mapping of aeromagnetic anomalies has identified substantially more. The basalts appear as distinct hillocks or mounds and outcrop with subvertical pillars. Contacts have only been observed in drill holes. Most of the basalts are situated in the Pre-Cambrian basement; only a few intrude Mesozoic sediments. The area with basalts is situated partly within and partly west of the junction between the PZ and the FBZ (Fig. 1).

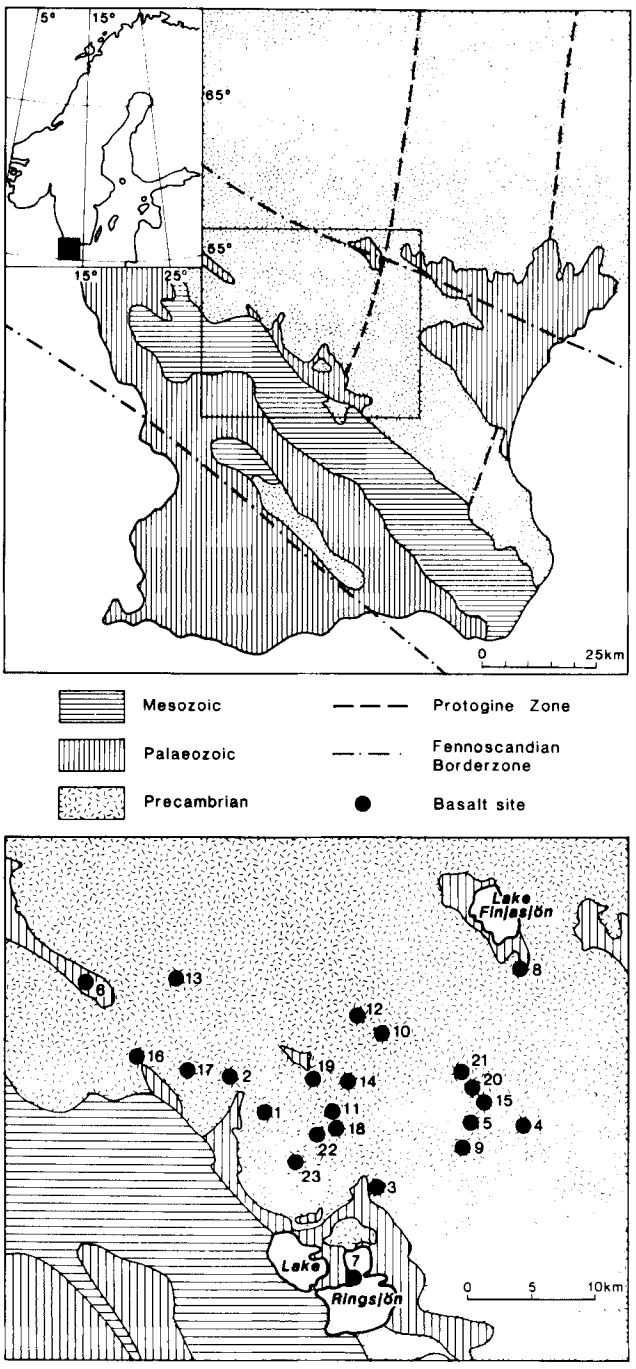


Figure 1. Geological sketch map of Scania, southern Sweden with palaeomagnetic sites.

AGE RELATIONS

Tuffs and lavas probably originating from the Scanian volcanoes have been found in drill holes where they cover Liassic sediments (Bölau 1966), and microfossils from a basaltic tuff have been dated to the upper part of the Early Jurassic: Toarcian to Aalenian (Tralau 1973). This constrains the age of the basalts as post-Rhaetian, probably Toarcian to Aalenian, or 194–181 Ma (Harland *et al.* 1989). In contrast, K–Ar age estimates of the basalts (Table 1) range from 171 to 111 Ma (Klingspor 1976) and 134 to 81 Ma (Printzlau & Larsen 1972). The latter authors considered some ages as too young due to loss of argon during devitrification. The K–Ar dating suggests that the volcanic activity in Scania started in the Middle Jurassic and continued intermittently into the Cretaceous, in which case it may be contemporary with post-Carboniferous, Jurassic to mid-Cretaceous activity in the North Sea basin between 175–165 Ma, 145–130 Ma and 110–95 Ma (Dixon, Fitton & Frost 1981).

PETROLOGY

The Scanian basalts are porhyric alkali basalts with phenocrysts of olivine and titanoaugite included in a groundmass of labradorite, titanoaugite, titanomagnetite and glass. Minor constituents are nepheline, biotite, sphene and picolite. Secondary alteration products such as zeolite, chlorite, epidote and calcite imply that the basalts have been affected by hydrothermal activities. They are locally enriched in xenoliths consisting of basement gneiss and gabbro and ultrabasic nodules ranging from pyroxenites to lherzolites and dunites. Initial ϵ_{Nd} values indicate a moderately depleted mantle as source for the basalts (Johannson 1990).

MAGNETIC MINERALOGY

Microscopic studies showed that the main opaque mineral is titanomagnetite in oxidation class 1 (Ade-Hall *et al.* 1968). It does not show any sign of high-temperature oxidation, except at site 11, where ilmenite lamellae were found in a few large grains. The grain size varies between 4 and 40 μm with the smaller grains prevailing. The ore content varies between 6 and 17 per cent (Norin 1933, 1934). In addition, minute grains of magnetite can be observed as a secondary mineral phase in olivine, augite and glass.

Strong-field thermomagnetic curves for 36 samples have been measured using a horizontal force balance. Typical behaviour is shown in Fig. 2 indicating mainly unoxidized titanomagnetite. The reversibility of the low Tc was investigated by heating the samples to the Curie temperature. The results showed that magnetic phases were reversible when heated to about 300 °C; at higher temperatures irreversible changes occurred. Primary unoxidized class 1 titanomagnetite normally has a Curie

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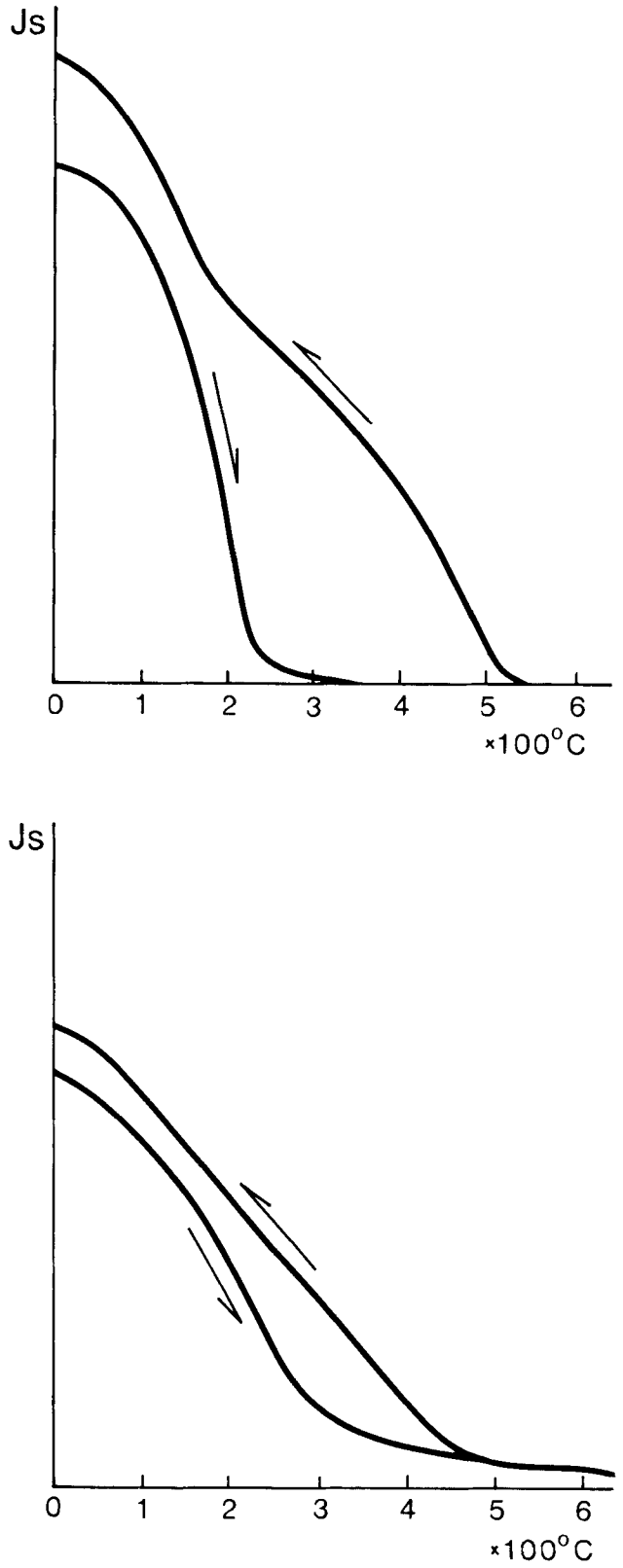


Figure 2. Examples of saturation magnetisation (*Js*) versus temperature. *Js* is given in arbitrary units.

temperature (Tc) below 300 °C depending on the Ti content. Tc's above 300 °C may imply that the basalts have suffered from low-temperature oxidation. Finely divided ferri-rutile granules are normally seen as a result of low-temperature

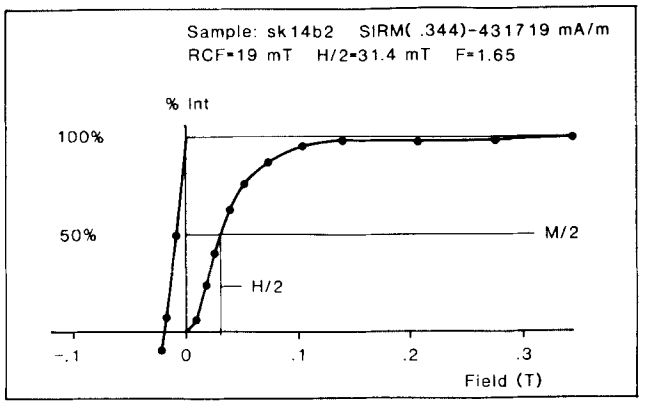


Figure 3. Saturation isothermal remanence. RCF = remanent coercive force, H = induced field and F = the ratio $H/2$ to RCF.

oxidation caused by hydrothermal activity. In the Scanian basalts such granules were observed. Similar observations were made by Ade-Hall *et al.* (1968) in a study of the Sky lavas. Ade-Hall, Palmer & Hubbarb (1971) noticed that an increasing degree of hydrothermal alteration causes a gradual change from low Tc to higher Tc.

Isothermal remanence (IRM) acquisition curves with remanence coercive force (RCF) from 15.3 mT to 23.8 mT and F values (remanent-acquisition coercive force/remanent coercive force) from 1.23 to 1.65 (Fig. 3), corresponding to titanomagnetite with varying Ti content (Dankers 1981) in accordance with the microscopic and thermomagnetic studies.

PALAEOMAGNETIC METHODS AND RESULTS

Sampling was carried out either by hand sampling or with a hand-held core drill. *In situ* orientation was made with a magnetic compass and, when possible, a sun compass. The Natural Remanent Magnetism (NRM) was measured on Digico or Molspin fluxgate spinner magnetometers and a 85 Hz JR3 spinner magnetometer. Conventional Alternating Field (AF) and thermal demagnetization techniques were applied to isolate the characteristic NRM. The characteristic remanence components were calculated by a least-square method (Torsvik 1986). Site mean directions were calculated according to Fisher (1953) and the results are summarized in Table 1. The palaeomagnetic studies were performed in the laboratories of Lund, Bergen and Sogndal and samples from some sites were studied at all laboratories. The respective results are given in Table 1. Also included are the data from the study made by Schweitzer (1975). That means that samples from site 9 were studied independently at three laboratories and samples from a few other sites by two laboratories (Table 1). One site, no. 2 in Fig. 1, did not yield any reliable data and is omitted from Table 1.

The results of the AF demagnetization showed that sites with well grouped NRM directions, of both polarities, did not reveal any large directional changes during demagnetization to 100 mT. In most cases it was not possible to demagnetize the samples at higher fields because viscous components made further measurements meaningless. Figs 4(a) and (b) give examples of different types of behaviour during stepwise AF demagnetization.

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Table 1. N = number of samples, Decl. and Incl. are mean declination and inclination, α_{95} = 95 per cent confidence circle of mean remanence direction, k = Fisher's precision parameter, the age is given in Ma, P&L are data from Printzlau & Larsen (1972), K are data from Klingspor (1976), Lab. indicates if the measurements were made in the Sogn and Fjordane College lab. (S), In Lund (L) or in Munich (M) by Schweitzer (1975).

Site	N	Decl.	Incl.	α_{95}	k	Age	Lab
1. Allarpsberg	5	60	55	12.5	38	-	BS
3. Anneklev	6	49	45	11.5	35	95 +/- 5, P&L	L
4. Ballran	3	306	83	11.2	122	-	BS
5. Bolmarehus	7	32	67	3.5	279	129 +/- 6, P&L	M
6. Bonarp	7	199	-67	5.1	140	-	BS
"	6	185	-67	3.9	300	-	L
Mean Bonarp	2	192	-67	-	-	-	-
7. Bosjökloster	3	195	-60	8.5	214	-	BS
8. Espet	6	50	36	2.3	886	-	L
9. Göbnehall	7	237	-62	5.8	110	171 +/- 4, K	L
"	8	236	-57	6.1	82	134 +/- 8, P&L	BS
"	13	244	-54	5.2	63	"	M
Mean Göbneh	3	239	-58	4.7	298	"	-
10. Hagstadberg	5	304	-31	12.8	24	-	L
11. Hästhallarna	4	18	67	4.7	381	-	L
12. Holma	5	132	-73	6.3	150	-	L
13. Juskushall a	3	36	61	4.6	308	-	L
" b	4	92	-25	16.3	19	-	L
14. Knösen	5	49	72	7.7	99	-	L
15. Lönneberg	8	20	72	2.1	708	111 +/- 3, K	L
"	8	18	73	3.9	186	123 +/- 3, P&L	M
Mean Lönneb	2	19	73	-	-	-	-
16. Rallate	6	273	-55	3.9	291	111 +/- 3, K	L
17. Randslidern	6	353	73	4.6	210	-	L
18. Röan	7	10	73	5	147	-	M
19. Säte	5	311	-79	4.9	245	101 +/- 7, P&L	L
20. Snababerg	6	46	75	4.7	207	90 +/- 13, P&	L
21. Sösdala	12	34	77	2.8	240	-	L
22. Stenkilstorp	6	10	42	4.6	213	81 +/- 7, P&L	BS
23. Ulfsberg	18	32	61	5.9	35	-	L
Mean directions, sites 10 and 13b not included							
Positive polarity	15	33	66	7.4	24		
Negative polarity	6	225	-73	15.3	14		
Combined	21	35	68	6.8	20		

The directional results of the thermal demagnetizations were similar to those obtained during AF demagnetization [Figs 4(c) and (d)]. Blocking temperatures varied from 150 °C to 500 °C. Most samples showed major intensity drops in the interval 200 to 400 °C. It is important to notice that samples with different blocking temperatures revealed the same cleaned directions.

DISCUSSION

Site mean directions are listed in Table 1 and plotted in Fig. 5. The time involved in cooling through the blocking temperatures of the magnetic minerals in the small Scanian volcanic vents was probably less than the time needed to average out the geomagnetic secular variation, and quite large directions between site scatter can be expected. The presence of glass in the groundmass in most of the basalts sampled also indicates a rapid cooling. The expected scatter is also seen in Fig. 5. In site 13 two directions were observed, one well defined with steep positive inclination (a) in agreement with the mean basalt direction. The other direction (14b) with a shallow negative inclination does not fit the pattern. A shallow negative inclination direction, but with different declination, was found at site 10. The reason for these deviations are not clear, they may be attributed to secular variation or local tectonic movements.

The tectonics in the area are dominated by fault movements, and local tilting of minor blocks may have occurred and so increased the directional scatter between sites. Major rotational movements of the basalt area can be ruled out as the Carboniferous–Permian dolerites that transect the area give palaeomagnetic directions that are in good agreement with contemporary data from elsewhere in Scandinavia (Bylund 1974). Further, middle-to-late Proterozoic dolerites in the PZ part of the area give directions corresponding to directions from contemporary rocks in other parts of the PZ (Bylund 1992).

Since the Scanian palaeopole is derived from dual polarity data and the means of the normal, reversed and combined site mean directions are identical within the error limits, it seems justified to conclude that the two polarity magnetizations represent one magnetization component acquired at the time of the intrusion of the basalts.

The calculated palaeopole, with both polarities combined (Table 2), is situated at 69°N, 102°E, $dp = 10°$, $dm = 11°$. This pole position corresponds with the results from Toarcian sediments in France at 70°N, 103°E, A95 = 5.3° (Galbrun & Daly 1987) indicating a similar age for the magnetization of the Scanian basalts. Results from boreholes (Bölau 1966) and pollen studies from basaltic tuffs (Tralua 1973) constrain the age of the basalts as post-Rhaetian, probably Toarcian to Aalenian. This means 194–181 Ma, according to the time scale of Harland *et al.* 1989. The K–Ar age estimates range from 171 to 111 Ma (Klingspor 1976) and 134 to 81 Ma (Printzlau & Larsen 1972), but as secondary alterations in the basalt may have affected the K–Ar isotope system, these determinations probably represent minimum ages. Site 9 (Göbnehall) was dated by both laboratories, Klingspor (1976) reports an age of 171 ± 4 Ma while Printzlau & Larsen (1972) give an age estimate of 134 ± 8 Ma for the same basaltic neck. Due to the great uncertainty in the radiometric age estimates from the Scanian basalts by the K–Ar method, we suggest that the age estimates (194–181 Ma) taken from pollen from basaltic tuffs (Tralau 1973) are likely to be more reliable.

The results from the present study can be compared with contemporary North American palaeopoles after compensating for the opening of the Atlantic. The data of Frei & Cox (1987) are used. Their data take into account the relative displacement between Eurasia and North America prior to the formation of oceanic crust in the North Atlantic. The reconstruction is based on three independent data sets: sea-floor spreading (SF) data, shelf-contour lines and the APWP data. A rotation pole located at 88°N, 215°E with a rotation angle of −42° resulted in a well defined combined APWP for Eurasia and North America for the period 290–240 Ma is consistent with the geology of the continental margins of the North Atlantic. Frei & Cox (1987) suggest that the difference between their rotation pole and poles based on SF alone probably reflects relative motions of the two cratons in the time between the Early Triassic and Late

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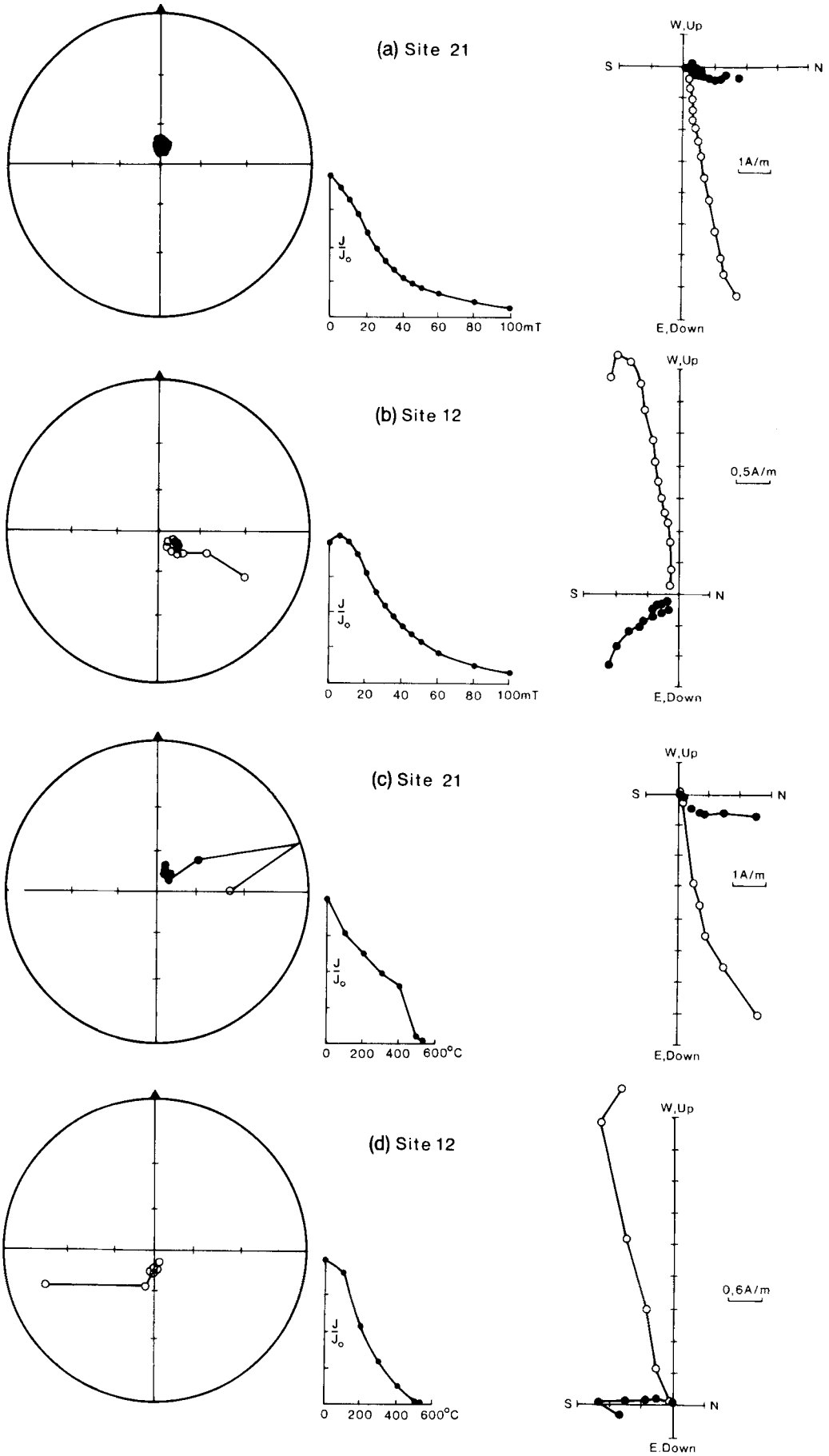


Figure 4. Behaviour of basalt samples to progressive alternating field (a and b) and thermal magnetic cleaning (b and c). Left: stereoplots, filled (open) circles denote positive (negative) inclination. Centre: intensity decay curves. Right: orthogonal projections of the end points of the NRM vectors during demagnetization, open circles plot in the vertical N–S plane, closed circles plot in the horizontal W–N plane.

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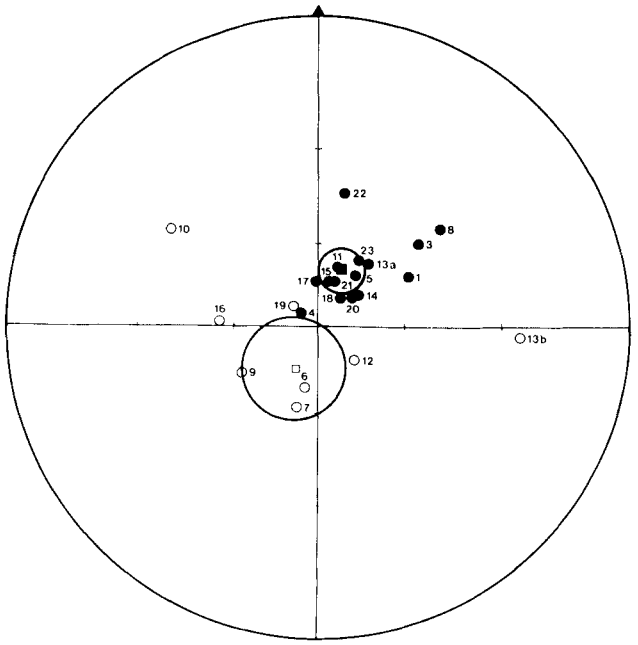


Figure 5. Site mean directions. Filled (open) circles denote positive (negative) inclination. The squares indicate mean of the site directions with α_{95} confidence circles.

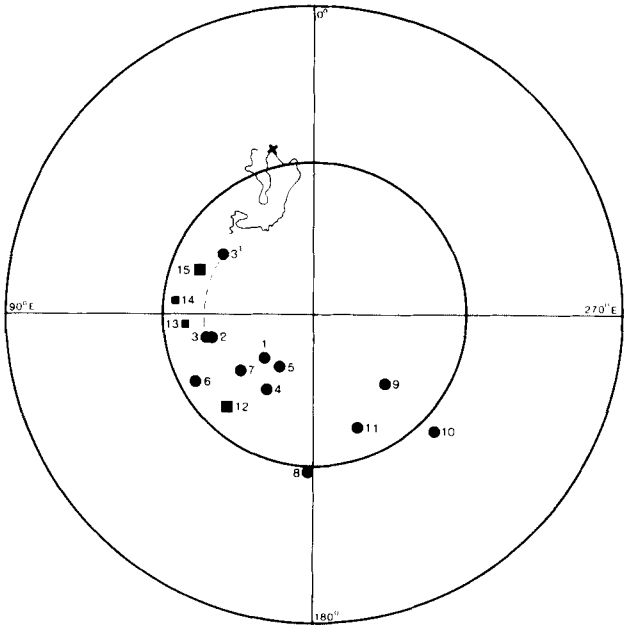


Figure 6. European and North American palaeopoles used for comparison (Table 2). The dashed line and pole 3′ indicates the position of the Scanian palaeopole after closing the Atlantic according to Frei & Cox 1987).

Cretaceous. Significant displacement probably took place in the North Atlantic prior to the formation of oceanic crust.

The position of the Scanian palaeopole after the rotation of the Eurasian APWP is 70°N, 57°E (Fig. 6). The rotation brings the Scanian pole close to the Early Jurassic East Rock Intrusion pole of Smith & Noltimier (1979) situated at 70°N, 67°E. The Scanian palaeopole lies, however, west of the majority of Early and Middle Jurassic palaeopoles (Fig. 6) for North America (van der Voo 1990). This discrepancy can be attributed to inaccuracies in the applied rotation data, in problems with the North American Early to Middle Jurassic palaeopoles or in incomplete removal of secondary magnetizations in the Scanian basalts. It is also possible that the 21 individual basalts investigated may be too few to give complete averaging of palaeosecular variation.

CONCLUSION

The Scanian palaeopole compares reasonably well with contemporary palaeopoles from North America after compensating for the opening of the Atlantic Ocean.

Due to the great uncertainty in the radiometric age estimates from the Scanian basalts by the K–Ar method, the age estimates (194–181 Ma) from borehole data (Bölau 1966) and pollen from basaltic tuffs (Tralau 1973) appear to be more reliable.

The palaeomagnetic data obtained imply that the Scanian basalts are of Toarcian age, c. 194 to 188 Ma and predate or are contemporary with the first period of Jurassic volcanic activity in the North Sea.

Table 2. The ages are adopted from Harland *et al.* 1989. A_{95} = 95 per cent confidence circles of mean pole positions.

No.	Rock formation	Stratigraphic age	Age, Ma	Pole position N	Pole position E	A95	References
1.	Yorkshire sed.	L.-M. Pliensbachian	200-195	77	135	2.5	Hijab and Tarling, 1982
2.	Toarcian, France	Toarcian	194-188	70	103	5.3	Galbrun and Daly, 1987
3.	Basalt. Scania	Toarcian-Aalenian	194-181	69	102	8.1	This paper
4.	Krakow-Cz. Upland	M.-U. Jurassic	188-144	72	150	7.3	Kruczyk et al., 1980
5.	Jura Mountains	Bajocian-Kimmeridgian	181-150	78	150	5.9	Johnson et al., 1984
6.	Alsace (lm+sb)	U. Bajocian	175-173	63	120	6.3	Kadzialko-Hoffmokl et al., 1984
7.	Malm, Limestone	M. Oxfordian-M. Kimmeridgia	160-153	69	128	4	Heller, 1977
8.	Isfjorden dol.	L. Jurassic-E. Cretaceous	-	58	179	9	Krumsiek et al., 1968
9.	Hornsund dol.	"	-	70	223	5	Vincenz et al., 1981
10.	Agardbukta dol.	"	-	54	225	9	Vincenz et al. 1984
11.	Hinlopen dol. comb.	"	144 +/-5	66	200	4.2	Halvorsen, 1990
12.	North American mean	L. Jurassic-L.M. Jurassic	176-145	65	136	8.4	van der Voo, 1990
13.	"	M.E. Jurassic-L. Jurassic	195-177	67	93	4.3	" "
14.	"	E. Jurassic-L. Triassic	215-196	62	82	4.7	" "
15.	East Rock intrusion	L. Jurassic	-	67	70	3.6	Smith and Noltimier, 1979

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